A Beginner's Guide to Mininet

You can instantly create realistic virtual networks deploying controllers, switches and hosts using Mininet. And experiment with them to your heart's content to run real kernel, application and switch code on a single machine, whether on a VM, the cloud or native.

Mininet is open source software that is used to simulate a software defined network (SDN) and its compatible controllers, switches and hosts. It comes in handy for networking enthusiasts and beginners to get some experience in working with the fast growing SDN technology. To put it simply, a SDN separates the control plane from the data forwarding plane, making network devices such as switches and routers fully programmable and, hence, the network behaves according to the users' requirements. By default, Mininet provides OVS switches and OVS controllers. However, it has the support to install other/preferred SDN controllers and switches instead of the defaults. The primary feature that distinguishes SDN devices from traditional network devices is the scope for customising protocols and functions.

Mininet supports the Openflow protocol, which provides an interface between the control plane and the data forwarding plane. Openflow protocols are used to control the packet flow as per the API written on the controller. Mininet also has support for a variety of topologies and ensures the availability of custom topologies. The CLI (command line interface) provided by Mininet is comfortable to use after a bit of practice.

Installing Mininet on your computer

To install Mininet, open the terminal and issue the following command:

```
# apt-get install mininet , to install the Mininet packages
on your system .
```

Next, verify the installation by issuing the following command:

```
# mn
```

After a successful installation, you'll see what's shown in Figure 1 on the screen.

Notice that a network is created with default topology (as shown in Figure 2) and default OVS switches. This network is ready for use and has all the parameters like IP addresses and links pre-configured, based on the default settings.

Getting started with the Mininet commands

The command to display the nodes present in the network is:

```
mininet> nodes
```